TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	79.1 ± 3.36
Correlation Stability	81.7 ± 2.97
Distribution Consistency	87.5 ± 2.87
Entropy Preservation	87.3 ± 1.68
Mean Z Score Drift	0.1 ± 0.04
Categorical Fidelity	93.1 ± 2.81
Privacy Risk	52.3 ± 1.19
Data Density	100.0 ± 0.01
Coverage Score	13.6 ± 3.88
Feature Accuracy	84.2 ± 5.40
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4.1-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	45.264	+0.518	1.2%	OK
TB	3.299	2.447	-0.852	25.8%	Drift
DB	1.486	1.114	-0.372	25.0%	Drift
Alkphos	290.576	327.210	+36.634	12.6%	Warn
Sgpt	80.714	74.897	-5.817	7.2%	OK
Sgot	109.911	79.436	-30.475	27.7%	Drift
TP	6.483	6.733	+0.250	3.9%	OK
ALB	3.142	3.296	+0.154	4.9%	OK
A/G Ratio	0.947	0.949	+0.002	0.3%	OK
Selector	1.286	1.300	+0.014	1.1%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	79.1 ± 3.36
Correlation Stability (*2)	81.7 ± 2.97
Distribution Consistency (*3)	87.5 ± 2.87
Entropy Preservation (*4)	87.3 ± 1.68
Mean Z Score Drift (*5)	0.1 ± 0.04
Categorical Fidelity (*6)	93.1 ± 2.81
Privacy Risk (*7)	52.3 ± 1.19
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	13.6 ± 3.88

Feature Accuracy (*10)	84.2 ± 5.40
Novelty Score (*11)	0.4 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6991452991452991	0.6589743589743591	0.6905982905982906	-0.008547008547008562	
F1	0.8034012629655765	0.7543508086519378	0.7840698279700442	-0.019331434995532238	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6752136752136753	0.6820512820512821	-0.00427350427350428	
F1	0.7971299319768493	0.7617216748368555	0.7838217973148851	-0.013308134661964299	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.6606837606837608	0.7017094017094017	-0.0017094017094016923	
F1	0.8108734553482932	0.7520205407010907	0.7985624494119328	-0.012311005936360286	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.6752136752136753	0.6085470085470086	0.04957264957264958	*
F1	0.5587432119380917	0.7538436409378921	0.6363783943765997	0.07763518243850812	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6632478632478633	0.6094017094017096	0.6760683760683761	0.01282051282051282	
F1	0.764888119133815	0.7104863663738259	0.7705660871594265	0.005677968025611458	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.664957264957265	0.6897435897435897	-0.0170940170940171	
F1	0.8279091624529537	0.7506710307196236	0.7942040036904237	-0.03370515876253004	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6811965811965812	0.6606837606837608	0.7076923076923076	0.02649572649572649	
F1	0.7886776692401435	0.7584998629415364	0.8022077287586518	0.01353005951850836	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6846153846153846	0.7153846153846153	0.01282051282051280	
F1	0.7997903571549948	0.7792100229618588	0.8060409949513447	0.00625063779635002	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7008547008547008	0.6700854700854701	0.7128205128205127	0.01196581196581196	
F1	0.8069166269431687	0.7689999491181714	0.8105812921675449	0.00366466522437599 58	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.676923076923077	0.6854700854700855	0.01111111111111117	
F1	0.7821343746683586	0.7749485263683191	0.7864708305415334	0.00433645587317482 7	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Male	2.5	1.2	163	28	22	7.6	4.0	1.1
2	39	Male	6.6	3.0	215	190	950	4.0	1.7	0.7
3	27	Male	0.6	0.2	161	27	28	3.7	1.6	0.76
4	57	Male	1.0	0.3	187	19	23	5.2	2.9	1.2
5	30	Female	0.8	0.2	158	25	22	7.9	4.5	1.3
6	66	Female	2.9	1.3	168	21	38	5.5	1.8	0.4
7	34	Male	1.2	0.5	245	45	38	6.8	3.4	1.0
8	29	Female	0.7	0.3	160	22	20	7.2	4.2	1.3
9	52	Male	3.5	1.6	520	120	110	6.0	2.8	0.7
10	41	Male	0.9	0.4	180	30	28	7.1	3.7	1.1
11	63	Male	7.8	3.9	680	210	190	5.5	2.2	0.5
12	27	Female	0.8	0.2	150	18	15	7.5	4.5	1.4
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	2.7	1.3	219	60	180	7.0	3.2	0.8
2	40	Male	1.2	0.6	204	23	27	7.6	4.0	1.1
3	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
4	52	Male	0.6	0.1	171	22	16	6.6	3.6	1.2
5	40	Male	1.1	0.3	230	1630	960	4.9	2.8	1.3
6	57	Male	4.0	1.9	190	45	111	5.2	1.5	0.4
7	45	Male	1.5	0.7	210	45	40	6.8	3.2	1.0
8	38	Female	0.8	0.3	180	30	25	7.2	4.0	1.2
9	60	Male	3.2	1.5	350	55	60	6.0	2.8	0.7
10	29	Female	0.7	0.2	160	25	20	7.5	4.3	1.1
11	55	Male	2.8	1.3	300	40	50	6.5	3.0	0.9
12	42	Male	1.0	0.4	220	35	30	7.0	3.5	1.1
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	29	Female	0.7	0.1	162	52	41	5.2	2.5	0.9
2	60	Male	2.0	0.8	190	45	40	6.0	2.8	0.8
3	32	Male	15.6	9.5	134	54	125	5.6	4.0	2.5
4	50	Male	0.9	0.2	202	20	26	7.2	4.5	1.66
5	38	Male	1.5	0.4	298	60	103	6.0	3.0	1.0
6	35	Male	1.6	0.7	157	15	44	5.2	2.5	0.9
7	34	Male	1.2	0.5	210	45	50	6.5	3.2	0.9
8	47	Male	0.8	0.3	180	30	28	7.0	3.6	1.1
9	29	Female	0.7	0.2	160	20	18	6.8	3.4	1.0
10	58	Male	3.5	1.8	450	90	85	6.0	2.8	0.7
11	42	Male	0.9	0.4	220	35	40	7.2	3.9	1.2
12	25	Male	0.6	0.1	140	15	20	7.5	4.0	1.0
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	40	Male	1.9	1.0	231	16	55	4.3	1.6	0.6
2	37	Male	0.8	0.2	147	27	46	5.0	2.5	1.0
3	42	Female	7.4	3.6	298	52	102	4.6	1.9	0.7
4	40	Male	0.6	0.1	98	35	31	6.0	3.2	1.1
5	27	Male	1.2	0.4	179	63	39	6.1	3.3	1.1
6	31	Male	0.6	0.1	175	48	34	6.0	3.7	1.6
7	35	Male	1.1	0.4	210	45	40	6.5	3.2	0.9
8	42	Female	0.8	0.2	190	20	18	7.0	3.5	1.0
9	29	Male	0.7	0.2	160	30	25	6.8	3.7	1.1
10	50	Male	2.5	1.0	300	60	55	6.0	2.8	0.7
11	38	Female	0.9	0.3	180	25	20	7.2	3.6	1.2
12	45	Male	1.3	0.5	250	50	45	6.3	3.0	0.8
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	36	Male	0.8	0.2	158	29	39	6.0	2.2	0.5
2	72	Male	0.6	0.1	102	31	35	6.3	3.2	1.0
3	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4
4	42	Female	7.4	3.6	298	52	102	4.6	1.9	0.7
5	70	Male	0.6	0.1	862	76	180	6.3	2.7	0.75
6	42	Female	0.9	0.2	165	26	29	8.5	4.4	1.0
7	45	Male	1.2	0.5	210	40	35	6.5	3.5	1.0
8	38	Female	0.8	0.3	180	25	22	7.2	4.0	1.2
9	60	Male	3.5	1.7	450	55	60	5.8	2.8	0.7
10	29	Male	0.9	0.4	190	30	28	7.0	3.9	1.1
11	52	Female	0.7	0.2	170	20	18	7.5	4.2	1.3
12	67	Male	2.8	1.3	300	45	50	6.0	3.0	0.8
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	35	Female	0.9	0.3	158	20	16	8.0	4.0	1.0
2	35	Male	0.7	0.2	198	42	30	6.8	3.4	1.0
3	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
4	39	Male	1.9	0.9	180	42	62	7.4	4.3	1.38
5	63	Male	0.5	0.1	170	21	28	5.5	2.5	0.8
6	60	Male	5.8	2.7	599	43	66	5.4	1.8	0.5
7	34	Male	1.1	0.4	210	35	40	6.8	3.3	1.0
8	45	Female	0.7	0.2	160	20	18	7.2	3.8	1.2
9	29	Male	0.9	0.3	180	45	50	6.5	3.5	0.9
10	52	Male	2.5	1.1	400	80	90	6.0	2.5	0.6
11	38	Female	0.8	0.3	140	25	20	7.4	4.0	1.1
12	61	Male	5.0	2.5	600	120	130	5.5	2.0	0.5
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	34	Male	8.7	4.0	298	58	138	5.8	2.4	0.7
2	48	Female	1.1	0.7	527	178	250	8.0	4.2	1.1
3	55	Male	14.1	7.6	750	35	63	5.0	1.6	0.47
4	20	Male	1.1	0.5	128	20	30	3.9	1.9	0.95
5	6	Male	0.6	0.1	289	38	30	4.8	2.0	0.7
6	65	Male	1.9	0.8	170	36	43	3.8	1.4	0.58
7	37	Male	1.5	0.7	210	45	40	6.5	3.2	1.0
8	29	Female	0.9	0.3	180	22	25	7.8	4.1	1.2
9	54	Male	3.8	1.9	320	65	70	5.9	2.5	0.7
10	48	Male	2.1	1.0	275	38	42	6.2	3.0	0.9
11	62	Female	1.2	0.5	160	20	18	7.0	3.6	1.1
12	45	Male	0.8	0.3	190	30	28	6.8	3.7	1.0
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	31	Male	1.3	0.5	184	29	32	6.8	3.4	1.0
2	75	Female	0.8	0.2	205	27	24	4.4	2.0	0.8
3	62	Male	0.7	0.2	173	46	47	7.3	4.1	1.2
4	42	Female	0.8	0.2	195	18	15	6.7	3.0	0.8
5	34	Male	4.1	2.0	289	875	731	5.0	2.7	1.1
6	32	Male	18.0	8.2	298	1250	1050	5.4	2.6	0.9
7	44	Male	0.9	0.3	210	38	29	6.7	3.2	1.0
8	29	Female	0.8	0.2	190	22	18	7.1	3.7	1.1
9	53	Male	1.2	0.5	250	45	40	6.5	3.0	0.8
10	37	Male	0.7	0.1	180	25	20	7.3	4.1	1.2
11	61	Female	1.0	0.4	300	30	35	6.2	2.8	0.9
12	48	Male	0.8	0.2	220	40	38	6.8	3.3	1.0
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	29	Male	0.7	0.2	165	55	87	7.5	4.6	1.58
2	60	Male	2.6	1.2	171	42	37	5.4	2.7	1.0
3	34	Male	5.9	2.5	290	45	233	5.6	2.7	0.9
4	33	Male	0.7	0.1	168	35	33	7.0	3.7	1.1
5	8	Female	0.9	0.2	401	25	58	7.5	3.4	0.8
6	75	Male	14.8	9.0	1020	71	42	5.3	2.2	0.7
7	34	Male	1.2	0.5	210	48	52	6.8	3.4	1.0
8	45	Female	0.7	0.2	180	35	30	7.2	4.0	1.1
9	60	Male	3.5	1.7	400	60	55	5.5	2.5	0.7
10	29	Male	0.8	0.3	190	25	28	7.0	3.8	1.3
11	52	Male	15.0	7.5	350	120	140	6.0	2.2	0.6
12	40	Female	0.9	0.3	220	40	38	7.5	3.9	1.0
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	19	Female	0.7	0.2	186	166	397	5.5	3.0	1.2
2	32	Male	12.1	6.0	515	48	92	6.6	2.4	0.5
3	22	Female	2.2	1.0	215	159	51	5.5	2.5	0.8
4	45	Male	0.8	0.2	140	24	20	6.3	3.2	1.0
5	21	Male	0.8	0.2	183	33	57	6.8	3.5	1.0
6	61	Male	1.5	0.6	196	61	85	6.7	3.8	1.3
7	34	Male	1.2	0.5	210	45	50	6.5	3.3	1.0
8	47	Female	0.9	0.3	185	30	28	7.1	3.7	1.1
9	29	Male	0.7	0.2	160	22	20	6.8	3.5	1.2
10	56	Male	2.5	1.1	320	75	80	5.9	2.8	0.7
11	41	Female	0.8	0.3	190	40	35	7.3	3.9	1.0
12	62	Male	1.0	0.4	150	25	30	6.2	3.1	0.9
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3